ASSIGNMENT-1

IMPORTANT INSTRUCTIONS

- *All questions in this assignment are coding questions.*

SUBMISSION FORMAT:

- *Create ONE Jupyter Notebook (.ipynb) file*
- *Solve all four questions in this single notebook.*
- *Add clear markdown cells with question titles.*
- *Include comments in your code to explain it wherever needed.*
- *Upload the completed (.ipynb) file to Google Classroom*

For queries, contact us on Whatsapp.

➢ Question 1: Ideal Gas Law and Isotherms

Part (a):

Create a function that takes temperature (in Kelvin) and volume (in m^3) as input and returns pressure (in Pa) as output. Use the ideal gas law: $PV = nRT$, where $n = 1$ mole and $R = 8.314$ J/(mol·K).

Note: All units must be in the SI (metric) system.

Part (b):

Create an array of volumes containing natural numbers from 1 to 50 (representing m^3). Using the function from part (a), plot two isotherms (temperature = 273 K and temperature = 323 K) on the same graph. The x-axis should represent volume (V in m^3) and the y-axis should represent pressure (P in Pa).

Requirements: Include axis labels, a title, and a legend to identify each isotherm.

➢ Question 2: Understanding Data Distribution with Random Arrays

Part (a):

Create three different random arrays using a uniform distribution between -100 and +100:

- **(i)** Array with **10 elements**
- **(ii)** Array with **100 elements**
- **(iii)** Array with **1000 elements**

Part (b):

For each of the three arrays, plot a histogram by dividing the data into 1000 equal intervals (bins). You should have three separate histograms.

Part (c):

Write an explanation of the results from part (b) in a **Markdown cell**. Discuss:

- How the shape of the histogram changes as the number of data points increases

- Why a uniform distribution should look flat (or nearly flat) across all bins
- How randomness affects the appearance of small vs. large datasets

➢ Question 3: Variable Thermal Conductivity (Class-based Approach)

Part (a):
Assume a hypothetical rod where the specific thermal conductivity varies with position according to: $k(x) = ax + d$, where $a = 1$, $d = 1$, and both are greater than zero.

Create a Python class called `ThermalRod` that:

- Stores the values of a and d in the `__init__` method
- Has a method called `get_temperature(x)` that takes position x (in meters) as input and returns the temperature $T(x)$ as output
- Assumes the boundary condition: $T(0) = 0$ K (temperature at the left end is 0)
- Uses numerical integration (via SciPy) to calculate the temperature at any position

Hint: The temperature at any point equals the total temperature difference multiplied by the ratio of thermal resistance between that point and the coldest point to the total thermal resistance of the rod.

Part (b):
Generate x-values from 0 to 1 meter with a resolution of 1 cm (0.01 m) and plot the **Temperature vs Position** graph using the class method from part (a).

Given Parameters:

- Length of rod = 1 m
- $a = 1$
- $d = 1$

- ### Question 4: Heat Transfer and Phase Change (Calorimetry)

Part (a):

Create a dictionary where:

- **Keys** = Material names (e.g., 'water', 'iron', 'copper', 'aluminum')
- **Values** = Specific heat capacities in J/(kg·K)

Use the following data:

- Water: 4180 J/(kg·K)
- Iron: 483 J/(kg·K)
- Copper: 399 J/(kg·K)
- Aluminum: 882 J/(kg·K)

Part (b):

Create a function that simulates the mixing of hot water with ice. The function takes:

- **M** = mass of hot water (in kg)
- **T** = initial temperature of water (in °C)
- **m** = mass of ice (in kg) at the freezing point (0°C)

The function should return:

- **If ice does NOT completely melt:** The mass of ice that melted (in kg)
- **If ice does completely melt:** The final equilibrium temperature (in °C) of the mixture

Given Parameters:

- Latent heat of fusion of ice = 334 kJ/kg = 334,000 J/kg
- Specific heat of water = 4180 J/(kg·K)

- Neglect the heat capacity of the tumbler (container)

Physics Formula to Use:

$$Q_{\text{lost by water}} = Q_{\text{gained by ice}}$$

Where:

- Heat lost by water = $M \cdot c_{\text{water}} \cdot (T - T_f)$
- Heat to melt ice = $m_{\text{melted}} \times L_f$

END Of Assignment-1